

Pesticide Application Checklist

When planning for possible pesticide applications in an IPM program, review and complete this checklist to consider practices that minimize environmental and efficacy problems.

Choose a pesticide from the UC IPM Pest Management Guidelines for the target pest considering:

- [Impact on natural enemies and honey bees\(/agriculture/potato/Relative-Toxicities-of-Insecticides-and-Miticides-Used-in-Potatoes-to-Natural-Enemies-and-Honey-Bees/\)](#). For more information, see [Protecting Natural Enemies and Pollinators\(/mitigation/protect_beneficials.html\)](#)
- Potential for water quality problems using the [UC IPM WaterTox database\(/TOX/simplewatertox.html\)](#).
- Impact on aquatic invertebrates. For more information, see [Pesticide Choice,\(https://anrcatalog.ucanr.edu/pdf/8161.pdf\)](#)UC ANR Publication 8161 (PDF).
- Chemical mode of action if pesticide resistance is an issue. For more information, see [Herbicide Resistance: Definition and Management Strategies\(http://anrcatalog.ucanr.edu/pdf/8012.pdf\)](#) UC ANR Publication 8012 (PDF).
- Endangered species that may be near your site. Find out using the Department of Pesticide Regulation's [PRESCRIBE\(http://cdpr.ca.gov/docs/endspec/prescint.htm\)](#) program.

Before an application

- Ensure that spray equipment is [properly calibrated\(/training/incorporating-calibration.html\)](#) to deliver the desired pesticide amount for optimal coverage.
- Use appropriate spray nozzles and pressure to minimize off-site movement of pesticides.
- Avoid spraying during these conditions to avoid off-site movement of pesticides.
 - Wind speed under 3 mph and over 10 mph
 - Temperature inversions
 - Just prior to rain or irrigation (unless it is an appropriate amount, such as when incorporating a soil-applied pesticide)
 - At tractor speeds over 2 mph
- Identify and take special care to protect sensitive areas (for example, waterways or riparian areas) surrounding your application site. Pay attention to buffer zones.
- Review and follow labeling for pesticide handling, personal protection equipment (PPE) requirements, storage, and disposal guidelines.
- Check and follow restricted entry intervals (REI) and preharvest intervals (PHI).

After an application

- Record application date, product used, rate, and location of application.
- Follow up to confirm that treatment was effective.

Consider water management practices that reduce pesticide movement off-site.

- Consult relevant publications: [Protecting Surface Water from Sediment-Associated Pesticides in Furrow-Irrigated Crops\(http://anrcatalog.ucanr.edu/pdf/8403.pdf\)](#), UC ANR Publication 8403 (PDF).

- Consult the [Department of Pesticide Regulation Groundwater Protection Program \(GWPA\)](http://www.cdpr.ca.gov/docs/emon/grndwtr) (<http://www.cdpr.ca.gov/docs/emon/grndwtr>) website for pesticide information and mitigation measures.
- Install an [irrigation recirculation or storage and reuse system\(/mitigation/water_reuse.html\)](/mitigation/water_reuse.html). Redesign inlets into tailwater ditches to reduce erosion. For more information, see these publications:
 - [Reducing Runoff from Irrigated Lands: Tailwater Return Systems\(http://anrcatalog.ucanr.edu/pdf/8225.pdf\)](http://anrcatalog.ucanr.edu/pdf/8225.pdf), UC ANR Publication 8225 (PDF).
 - [Reducing Runoff from Irrigated Lands: Storing Runoff from Winter Rains\(http://anrcatalog.ucanr.edu/pdf/8211.pdf\)](http://anrcatalog.ucanr.edu/pdf/8211.pdf), UC ANR Publication 8211 (PDF).
- Use drip rather than sprinkler or flood irrigation.
- Limit irrigation to amount required using soil moisture monitoring and [evapotranspiration\(/WEATHER/#WEATHERDATA\)](/WEATHER/#WEATHERDATA) (ET).
- Consider [vegetative filter strips\(/mitigation/veg_filtering.html\)](/mitigation/veg_filtering.html) or ditches. (For more information, see [Vegetative Filter Strips\(http://anrcatalog.ucanr.edu/pdf/8195.pdf\)](http://anrcatalog.ucanr.edu/pdf/8195.pdf), UC ANR Publication 8195 (PDF)
- Use polyacrylamide (PAM) tablets in furrow irrigation systems to prevent off-site movement of sediments.

Consider management practices that reduce air quality problems.

- When possible, reduce [volatile organic compounds \(VOC\) emissions\(/mitigation/reducing_voc.html\)](/mitigation/reducing_voc.html) by decreasing the amount of pesticide applied, choosing low-emission management methods, and avoiding fumigants and emulsifiable concentrate (EC) formulations

For more about mitigating the effects of pesticides see the [Mitigation page\(/mitigation\)](/mitigation).



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